

Experiment 2

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1. Aim:

- a. To implement and analyze SQL SELECT queries using filtering, sorting, grouping, and aggregation concepts in PostgreSQL for efficient data retrieval and analytical reporting.

2. Objective:

- a. To retrieve specific data using filtering conditions
- b. To sort query results using single and multiple attributes
- c. To perform aggregation using grouping techniques
- d. To apply conditions on aggregated data
- e. To understand real-world analytical queries

3. Experiment Steps:

- a. Create table queries:
 - i. Create table customer_orders (order_id int primary key, customer_name varchar(30) not null, product varchar(50) not null, quantity int not null check (quantity > 0), price decimal(8, 2) not null check (price >= 0), order_date date not null);
- b. Insert queries:
 - i. Insert into customer_orders values (1023, 'amit sharma', 'laptop', 1, 55000.00, '2024-01-05');
- c. Filtering Data using conditions
 - i. Select * from customer_orders where price > 45000;
- d. Sorting Query Results:
 - i. Sorting in descending order:
Select order_id, customer_name as name, price as shopping_value from customer_orders order by shopping_value desc;

ii. Priority based sorting:

Select order_id, customer_name as name, price as shopping_value from customer_orders order by shopping_value desc, order_id;

e. Grouping data for aggregation:

i. Select product, count(product) as quantity_sold, sum(quantity * price) as total_sales from customer_orders group by product;

f. Applying conditions on aggregate data:

i. Select product, sum(quantity) as quantity_sold, sum(quantity * price) as total_sales from customer_orders group by product having sum(quantity) >= 5;

4. Output

a. Filtering data using conditions:

	order_id [PK] integer	customer_name character varying (30)	product character varying (50)	quantity integer	price numeric (8,2)	order_date date
1	1023	Amit Sharma	Laptop	1	55000.00	2024-01-05
2	4121	Priya Singh	Laptop	2	52000.00	2024-01-08
3	9187	Vikas Gupta	Laptop	1	58000.00	2024-01-13

b. Sorting in descending order:

	order_id [PK] integer	name character varying (30)	shopping_value numeric (8,2)
1	9187	Vikas Gupta	58000.00
2	1023	Amit Sharma	55000.00
3	4121	Priya Singh	52000.00
4	2678	Meena Joshi	12500.00
5	5876	Ankit Patel	12000.00

c. Priority based sorting:

	order_id [PK] integer	name character varying (30)	shopping_value numeric (8,2)
1	9187	Vikas Gupta	58000.00
2	1023	Amit Sharma	55000.00
3	4121	Priya Singh	52000.00
4	2678	Meena Joshi	12500.00
5	5791	Divya Kapoor	12000.00
6	5876	Ankit Patel	12000.00
7	6802	Manoj Kulkarni	12000.00

d. Grouping data for aggregation:

	product character varying (50) 🔒	quantity_sold bigint 🔒	total_sales numeric 🔒
1	Mouse	9	6650.00
2	Keyboard	9	12850.00
3	Monitor	7	83500.00
4	Laptop	4	217000.00

e. Conditions on aggregated data:

	product character varying (50) 🔒	quantity_sold bigint 🔒	total_sales numeric 🔒
1	Mouse	9	6650.00
2	Keyboard	9	12850.00
3	Monitor	7	83500.00

5. Learning Outcome

- Understand how data can be filtered to retrieve only relevant records from a database.
- Learn how sorting improves readability and usefulness of query results in reports.
- Gained the ability to group data for analytical purposes.
- Differentiate between row-level conditions and group-level conditions.
- Writing analytical SQL queries used in real-world scenarios.
- Better prepared to answer SQL-based questions related to filtering, grouping, and aggregation.